

HODGE THEORY I

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The purpose of this note is to give a heuristic dictionary between statements in l -adic cohomology and statements in Hodge theory. This dictionary has, among its sources, Serre's work [3] and Grothendieck's conjectural theory of motives [2]. Up to now it has mainly served to formulate conjectures in Hodge theory, and at times it has suggested proofs of them.

1.

Definition 1. A mixed Hodge structure H consists of:

- (a) a finitely generated $\mathbf{Z}$ -module $H_{\mathbf{Z}}$, the integral lattice;
- (b) a finite increasing filtration W on $H_{\mathbf{Q}} = H_{\mathbf{Z}} \otimes \mathbf{Q}$, the weight filtration;
- (c) a finite decreasing filtration F on $H_{\mathbf{C}} = H_{\mathbf{Z}} \otimes \mathbf{C}$, the Hodge filtration.

These data are subject to the following axiom: on $\mathrm{Gr}_W(H_{\mathbf{C}})$ there is a unique bigrading by subspaces $H^{p,q}$ such that

$$(i) \quad \mathrm{Gr}_W^n(H_{\mathbf{C}}) = \bigoplus_{p+q=n} H^{p,q};$$

- (ii) the filtration F induces on $\mathrm{Gr}_W(H_{\mathbf{C}})$ the filtration

$$F^p = \bigoplus_{p' \geq p} H^{p',q'};$$

- (iii) $H^{p,q} = \overline{H^{q,p}}$.

A morphism $f : H \rightarrow H'$ is a homomorphism $f : H_{\mathbf{Z}} \rightarrow H'_{\mathbf{Z}}$ such that $f : H_{\mathbf{Q}} \rightarrow H'_{\mathbf{Q}}$ and $f_{\mathbf{C}} : H_{\mathbf{C}} \rightarrow H'_{\mathbf{C}}$ are compatible respectively with the filtrations W and F .

The Hodge numbers of H are the integers

$$(1) \quad h^{p,q} = \dim_{\mathbf{C}} H^{p,q} = h^{q,p}.$$

We say that H is pure of weight n if $h^{p,q} = 0$ for $p + q \neq n$, equivalently if $\mathrm{Gr}_i^W(H) = 0$ for $i \neq n$. We also call such an object a Hodge structure of weight n .

The Tate Hodge structure $\mathbf{Z}(1)$ is the Hodge structure of weight -2 , purely of type $(-1, -1)$, for which $\mathbf{Z}(1)_{\mathbf{C}} = \mathbf{C}$ and

$$\mathbf{Z}(1)_{\mathbf{Z}} = 2\pi i \mathbf{Z} = \mathrm{Ker}(\exp : \mathbf{C} \rightarrow \mathbf{C}^*) \subset \mathbf{C}.$$

Put $\mathbf{Z}(n) = \mathbf{Z}(1)^{\otimes n}$.

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The mixed Hodge structures form an abelian category. If $f : H \rightarrow H'$ is a morphism, then $f_{\mathbf{Q}}$ and $f_{\mathbf{C}}$ are strictly compatible with the filtrations W and F [1, 2.3.5].

2.

Let A be a normal integral ring of finite type over $\mathbf{Z}$, let K be its field of fractions, and let $\overline{K}$ be an algebraic closure of K . Let K^{nr} be the largest subextension of $\overline{K}$ unramified at every prime ideal of A . One knows, or one defines,

$$\pi_1(\mathrm{Spec} A, \overline{K}) = \mathrm{Gal}(K^{nr}/K).$$

For every closed point x of $\mathrm{Spec} A$, defined by a maximal ideal m_x of A , the residue field $k_x = A/m_x$ is finite. The point x defines a conjugacy class of Frobenius substitutions $\varphi_x \in \pi_1(\mathrm{Spec} A, \overline{K})$. Put $q_x = \#k_x$ and $F_x = \varphi_x^{-1}$.

Let K be a field of finite type over the prime field of characteristic p , let $\overline{K}$ be an algebraic closure of K , let $l \neq p$ be prime, and let H be a finitely generated $\mathbf{Z}_l$ -module, or $\mathbf{Q}_l$ -vector space, endowed with a continuous action ρ of $\mathrm{Gal}(\overline{K}/K)$. We shall always suppose below that there exists A as above, with l invertible in A , such that ρ factors through

$$\pi_1(\mathrm{Spec} A, \overline{K}) = \mathrm{Gal}(K^{nr}/K).$$

We say that H is pure of weight n if, for every closed point x of a nonempty open subset of $\mathrm{Spec} A$, the eigenvalues α of F_x acting on H are algebraic integers all of whose complex conjugates have absolute value $q_x^{n/2}$.

Principle 2. *If the Galois module H comes from algebraic geometry, then on $H \otimes_{\mathbf{Z}_l} \mathbf{Q}_l$ there is a unique increasing filtration W , invariant under Galois, such that $\mathrm{Gr}_n^W(H)$ is pure of weight n .*

One may also expect $\mathrm{Gr}_n^W(H)$ to be semisimple.

When resolution of singularities is available, one can often give a conjectural definition of W , whose correctness follows from the Weil conjectures [5]; see section 6.

Let μ_{l^∞} be the subgroup of $\overline{K}^*$ formed by the roots of unity. The Tate module

$$\mathbf{Z}_l(1) = \mathrm{Hom}(\mathbf{Q}_l/\mathbf{Z}_l, \mu_{l^\infty})$$

is pure of weight -2 . Put $\mathbf{Z}_l(n) = \mathbf{Z}_l(1)^{\otimes n}$.

It is immediate that every morphism $f : H \rightarrow H'$ is strictly compatible with the weight filtration.

Principle 2 agrees with the fact that every extension of $\mathbf{G}_m$ (weight -2) by an abelian variety (weight $-1 > -2$) is trivial.

3. TRANSLATION

The Galois modules that occur in l -adic cohomology have, over $\mathbf{C}$, mixed Hodge structures as their analogues. We have moreover the following dictionary:

module pure of weight n	Hodge structure of weight n
weight filtration	weight filtration
Galois-compatible homomorphism	morphism
Tate module $\mathbf{Z}_l(1)$	Tate Hodge structure $\mathbf{Z}(1)$

4.

Let X be a complex algebraic variety, i.e. a scheme of finite type over $\mathbf{C}$, assumed separated. There is a subfield K of $\mathbf{C}$, of finite type over $\mathbf{Q}$, over which X can be defined; thus X is obtained by extension of scalars from a K -scheme X' . Let $\bar{K}$ be the algebraic closure of K in $\mathbf{C}$. Then $\text{Gal}(\bar{K}/K)$ acts on the l -adic cohomology groups $H^*(X_{\bar{K}}, \mathbf{Z}_l)$, and

$$H^*(X(\mathbf{C}), \mathbf{Z}) \otimes \mathbf{Z}_l = H^*(X, \mathbf{Z}_l) = H^*(X'_{\bar{K}}, \mathbf{Z}_l).$$

By section 3, one should expect the groups $H^n(X(\mathbf{C}), \mathbf{Z})$ to carry natural mixed Hodge structures. This can in fact be proved; see [1, 3.2.5] for the case in which X is smooth. The proof is algebraic, starting from classical Hodge theory [6]. For X projective and smooth, the Weil conjectures imply that $H^n(X_{\bar{K}}, \mathbf{Q}_l)$ is pure of weight n , while classical Hodge theory endows $H^n(X(\mathbf{C}), \mathbf{Z})$ with a Hodge structure of weight n . For every morphism $f : X \rightarrow Y$ and for K large enough, the pullback $f^* : H^*(Y_{\bar{K}}, \mathbf{Z}_l) \rightarrow H^*(X_{\bar{K}}, \mathbf{Z}_l)$ commutes with Galois. Likewise, $f^* : H^*(Y(\mathbf{C}), \mathbf{Z}) \rightarrow H^*(X(\mathbf{C}), \mathbf{Z})$ is a morphism of mixed Hodge structures.

For X smooth, the cohomology class in $H^{2n}(X_{\bar{K}}, \mathbf{Z}_l(n))$ of an algebraic cycle Z of codimension n , defined over K , is Galois invariant, i.e. it defines

$$c(Z) \in \text{Hom}_{\text{Gal}}(\mathbf{Z}_l(-n), H^{2n}(X_{\bar{K}}, \mathbf{Z}_l)).$$

Similarly, the cohomology class $c(Z) \in H^{2n}(X(\mathbf{C}), \mathbf{Z})$ is purely of type (n, n) , i.e. it corresponds to

$$c(Z) \in \text{Hom}_{\text{Hodge}}(\mathbf{Z}(-n), H^{2n}(X(\mathbf{C}), \mathbf{Z})).$$

5.

If $f : H \rightarrow H'$ is a Galois-compatible morphism between $\mathbf{Q}_l$ -vector spaces of different weights, then $f = 0$. Likewise, if $f : H \rightarrow H'$ is a morphism of pure mixed Hodge structures of different weights, then f is torsion. A more useful observation is the following.

Scholium 3. *Let H and H' be Hodge structures of weights n and n' with $n > n'$. Let $f : H_{\mathbf{Q}} \rightarrow H'_{\mathbf{Q}}$ be a homomorphism such that $f_{\mathbf{C}} : H_{\mathbf{C}} \rightarrow H'_{\mathbf{C}}$ respects F . Then $f = 0$.*

6.

Let $\overline{X}$ be a smooth projective variety over $\mathbf{C}$, let $D = \bigcup D_i$ be a normal crossings divisor in $\overline{X}$, a sum of smooth divisors, and let j be the inclusion in $\overline{X}$ of $U = \overline{X} - D$. For $Q \subset [1, n]$, put $D_Q = \bigcap_{i \in Q} D_i$.

In l -adic cohomology we have canonically

$$(2) \quad \mathcal{R}^q j_* \mathbf{Z}_l = \bigoplus_{|Q|=q} (i_Q)_* \mathbf{Z}_l(-q),$$

and the Leray spectral sequence for j is

$$(3) \quad {}'E_2^{p,q} = \bigoplus_{|Q|=q} H^p(D_Q, \mathbf{Q}_l) \otimes \mathbf{Z}_l(-q) \Rightarrow H^{p+q}(U, \mathbf{Q}_l).$$

By the Weil conjectures [5], $H^p(D_Q, \mathbf{Q}_l)$ is pure of weight p , hence $'E_2^{p,q}$ is pure of weight $p + 2q$. Being a quotient of a subobject of $'E_2^{p,q}$, $'E_r^{p,q}$ is again pure of weight $p + 2q$. By Scholium 3, $d_r = 0$ for $r \geq 3$, since the weights $p + 2q$ and $p + 2q - r + 2$ of $'E_r^{p,q}$ and $'E_r^{p+r, q-r+1}$ are different. Thus $'E_3^{p,q} = 'E_\infty^{p,q}$. Up to a reindexing, the weight filtration of $H^*(U, \mathbf{Q}_l)$ is the abutment of (3):

$$(4) \quad \mathrm{Gr}_n^W(H^k(U, \mathbf{Q}_l)) = 'E_2^{2k-n, n-k}.$$

7.

For the usual topology and integral cohomology, the Leray spectral sequence for j is

$$(5) \quad {}'E_2^{p,q} = \bigoplus_{|Q|=q} H^p(D_Q, \mathbf{Z}) \Rightarrow H^{p+q}(U, \mathbf{Z}).$$

Since each D_Q is a nonsingular projective variety, $'E_2^{p,q}$ carries a Hodge structure of weight p . Put $E_2^{p,q} = {}'E_2^{p,q} \otimes \mathbf{Z}(-q)$, a Hodge structure of weight $p + 2q$. As an abelian group, $E_2^{p,q} = {}'E_2^{p,q}$; it is useful to regard (5) as a spectral sequence with initial term $E_2^{p,q}$. By section 3, one expects $d_2 : E_2^{p,q} \rightarrow E_2^{p+2, q-1}$ to be a morphism of Hodge structures. This is proved by interpreting d_2 as a Gysin morphism. Thus $E_3^{p,q}$ has a Hodge structure of weight $p + 2q$. Again by section 3, one expects that, modulo torsion, the spectral sequence (3) degenerates at E_3 and that the vanishing of d_r for $r \geq 3$ is an application of Scholium 3. This program is carried out in [1, 3.2]. The weight filtration of $H^*(U, \mathbf{Q})$ is defined there as the abutment of (5), up to the reindexing (4).

In fact, in order to put mixed Hodge structures on cohomology groups H^* , the key point so far has always been to find a spectral sequence E abutting to H^* such that the l -adic analogue of $E_2^{p,q}$ is conjecturally pure, of weight $p + 2q$. Then $E_2^{p,q}$ should carry a natural Hodge structure of weight $p + 2q$, and the filtration W is the abutment of E .

8.

Let $\text{Spec}(V)$ be the spectrum of a Henselian discrete valuation ring, with fraction field K and residue field k of finite type over the prime field of characteristic p . Let $\bar{K}$ be an algebraic closure of K , and let H be a finite-dimensional $\mathbf{Q}_l$ -vector space, $l \neq p$, on which $\text{Gal}(\bar{K}/K)$ acts continuously. By Grothendieck [4, Appendix], a finite-index subgroup of the inertia group I acts unipotently. After replacing V by a finite extension, we reduce to the case where all of I acts unipotently, the semistable case. The action then factors through the largest pro- l quotient I_l of I , canonically isomorphic to $\mathbf{Z}_l(1)$.

Principle 4. *In the semistable case, if the Galois module H comes from algebraic geometry, then there is a unique increasing filtration W of H , the weight filtration, such that I acts trivially on $\text{Gr}_n^W(H)$ and such that $\text{Gr}_n^W(H)$, as a Galois module under $\text{Gal}(\bar{k}/k) \subset \text{Gal}(\bar{K}/K)/I$, is pure of weight n .*

Compare this with Principle 2 and with the appendix to [4]. When resolution of singularities is available, one can sometimes give a conjectural definition of W whose validity follows from the Weil conjectures. Using resolution and Weil, it is often easy to show at least that H can be broken up into Galois modules, under $\text{Gal}(\bar{k}/k)$, that are pure.

Assume that H is semistable. For $T \in I$, define $\log T$ by

$$-\sum_{n>0} (\text{Id} - T)^n / n.$$

The map $(T, x) \mapsto \log T(x)$ identifies with a homomorphism

$$(6) \quad M : \mathbf{Z}_l(1) \otimes_{\mathbf{Z}_l} H \rightarrow H.$$

Since $\mathbf{Z}_l(1)$ has weight -2 , necessarily

$$(7) \quad M(\mathbf{Z}_l(1) \otimes W_n(H)) \subset W_{n-2}(H),$$

and M induces

$$(8) \quad \text{Gr}_n(M) : \mathbf{Z}_l(1) \otimes \text{Gr}_n^W(H) \rightarrow \text{Gr}_{n-2}^W(H).$$

Definition 5. If X is a nonsingular projective variety over an algebraically closed field k , define

$$L : \mathbf{Z}_l(-1) \otimes H^*(X, \mathbf{Z}_l) \rightarrow H^*(X, \mathbf{Z}_l)$$

to be cup product with the cohomology class of a hyperplane section.

There is a formal analogy between L and M . Just as M is defined by an action of $\mathbf{Z}_l(1)$, one may regard L as defined by an action of $\mathbf{Z}_l(-1)$. The operator L raises degree by 2, and $\text{Gr}_n M$ in (8) lowers it by 2.

9.

Let D be the unit disk, $D^* = D - \{0\}$, and let

$$\begin{array}{ccc} X & \xrightarrow{\subset} & \mathbf{P}^r(\mathbf{C}) \times D \\ f \downarrow & & \downarrow \text{pr}_2 \\ D & \xlongequal{\quad} & D \end{array}$$

be a family of projective varieties parametrized by D , with f proper and $f|_{D^*}$ smooth.

Keep the notation of section 8. In the analogy between a Henselian trait and a small neighborhood of 0 in the complex line, one has the following dictionary:

D	$\text{Spec}(V)$
D^*	$\text{Spec}(K)$
a universal cover $\tilde{D}^*$ of D^*	$\text{Spec}(\bar{K})$
$\pi_1(D^*)$	inertia group I
$\pi_1(D^*) = \mathbf{Z} \simeq \mathbf{Z}(1)$	$I_l = \mathbf{Z}_l(1)$
X	projective scheme over $\text{Spec}(V)$
$X^* = f^{-1}(D^*)$	X_K
$X_D = X \times_D \tilde{D}^*$	$X_{\bar{K}}$
$\mathcal{H}^i f_* \mathbf{Z} _{D^*}$	$H^i(X_{\bar{K}}, \mathbf{Z}_l)$
$H^i(X_t, \mathbf{Z})$	$H^i(X_{\bar{K}}, \mathbf{Z}_l)$

Note that X_D is homotopy equivalent to each fiber $X_t = f^{-1}(t)$ for $t \in D^*$. Thus $H^i(X_t, \mathbf{Z})$ again corresponds to $H^i(X_{\bar{K}}, \mathbf{Z}_l)$, and the action of I corresponds to the monodromy transformation T .

Here too, a finite-index subgroup of $\pi_1(D^*)$ acts unipotently on $H^i(X_t, \mathbf{Q}) = H^i(X_D, \mathbf{Q})$. Put ourselves in the semistable case, where all of $\pi_1(D^*)$ acts unipotently; this amounts to replacing D by a finite cover. Let T be the action of the canonical generator of $\pi_1(D^*)$.

By sections 3 and 8, one expects $H^i(X_t, \mathbf{Q}) \simeq H^i(X_D, \mathbf{Q})$ to be endowed with an increasing filtration W , the groups $\text{Gr}_n^W H^i(X_t, \mathbf{Q})$ to be endowed with Hodge structures of weight n , and

$$\log T(W_n) \subset W_{n-2},$$

with $\log T$ inducing a morphism of Hodge structures

$$M : \mathbf{Z}(-1) \otimes \text{Gr}_n^W(H^i) \rightarrow \text{Gr}_{n-2}^W(H^i).$$

One would like (6), not only (7) and (8), to have an analogue. In fact, for each tangent vector u to D at 0, one can define a mixed Hodge structure $\mathcal{H}_u$ on $H^i(X_t, \mathbf{Z})$. The filtration W and the Hodge structures on $\text{Gr}_n^W(\mathcal{H}_u)$ are independent of u , and the dependence of $\mathcal{H}_u$ on u is expressed simply in terms of T . In analogy with (6), one finds that, for every u , $\log T$ induces a

homomorphism of mixed Hodge structures

$$M : \mathbf{Z}(1) \otimes H^i(X, \mathbf{Z}) \rightarrow H^i(X, \mathbf{Z}).$$

Finally, the analogy with Definition 5 is not misleading, though here the assumption that $f|_{D^*}$ is proper and smooth is probably essential. One proves that

$$(\log T)^k : \mathrm{Gr}_{n+k}^W(H^n(X_t, \mathbf{Q})) \rightarrow \mathrm{Gr}_{n-k}^W(H^n(X_t, \mathbf{Q}))$$

is an isomorphism for every k ; see [6, IV, 6, corollary to Theorem 5]. This characterizes the filtration W . Up to now, an analogue of the Hodge positivity theorem [6, IV, 7, corollary to Theorem 7] is available only in very special cases. One hopes that the mixed structures $\mathcal{H}$ determine the asymptotic behavior, as $t \rightarrow 0$, of the family of pure structures $H^i(X_t, \mathbf{Z})$ for $t \in D^*$.

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